Help4CERN meeting Jan 13

- Zoom meetings – continue Tuesdays at 2pm cerntime?

- DIS 2026 deadline moved to Jan 26 – we send 2 or 3 abstracts?

  High energy, high luminosity muon-hadron collider at CERN
  WG6 Current Upgrades and Future Experiments (+ WG1 Structure Functions and Parton Densities)

  PDFs and low-x physics at LHμC?
  WG1 + WG2 (+ WG4?)

  LHμC as HET factory
  WG3 Electroweak Physics and Beyond the Standard Model

White Paper

Today, planning LHμC Physics Case

- Assume 250 and 500 fb^{-1} of data for 500 GeV and 1 TeV muons, respectively; and 10 fb^{-1} for initial QCD studies
- Overview LHeC physics case + discuss LHμC one using LHeC and FCC-*eh* reports (including papers for ESPPU!), and other published papers, to re-scale sensitivities/errors using production cross-sections:

Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(l\,p \to \nu_l\,H + \text{jet} + \text{X})$ [fb]	52	164	438	731
$\sigma(l\,p \to l\,H + \text{jet} + \text{X})$ [fb]	5.6	19	135	221
$\sigma(l\,p \to l\,t\bar{t} + \text{X})$ [pb]	0.48	2.24	?	?
$\sigma(l\,p \to \nu_l\,t + \text{X})$ [pb]	1.34	10.7	31	58

Table 1. Inclusive cross-sections for four major processes at the 20 and 50 GeV LHeC as well as 500 GeV and 1 TeV LHμC. In the LHeC cases an 80% electron beam polarisation is assumed [5].

White Paper

Contents

1 Introduction [Krzysztof, Saleh]

2 Physics case [Ali Can, Hamzeh]

2.1 Exploring QCD frontier [Anna, Daniel, Piotr?]

2.1.1 Non-linear QCD and PDFs

2.1.2 Precision α_S

2.1.3 ...

2.1.4 Nuclear effects and nPDFs

2.2 HET Factory

2.2.1 Higgs case [Simone?]

2.2.2 Precision electroweak physics

2.2.3 Top factory

2.3 Unique BSM searches [ ]

2.3.1 ...

2.4 Neutrino physics [ ]

2.5 Outlook/next steps

3 High energy, high luminosity lepton-hadron colliders [Burak, Frank, Krzysztof, Umit]

3.1 LHeC

3.1.1 Stage-one LHeC

3.1.2 Stand-alone LHeC

3.2 LHμC

3.2.1 Ultra cold μ^+-source

3.2.2 μ^+ booster (two scenarios)

3.2.3 μ^+ main ring

3.2.4 Proton ring performance

3.2.5 Interaction region/MDI

3.2.6 Neutrino flux

3.2.7 1 TeV upgrade

3.2.8 LHμC performance

3.2.9 FCC-μh case

3.2.10 Outlook/next steps

4 Detectors [Ilhan, Laurent]